SEMTECH UNIFIED SOFTWARE PLATFORM (USP)

Radio Test CLI
User Manual

Interactive RF Test Interface for Semtech Radio Modules

SX126x • LR11xx • LR20xx

Version: v0.5.0

Platforms: NUCLEO-L476RG · NUCLEO-L073RZ · FPB-RA0E2 · Linux (Raspberry Pi) · Zephyr (nRF54L15)

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PART 1

Quick Start

Build the CLI, connect your radio, and run your first test.

1.1 Requirements

Semtech radio module: LR1110, LR1120, LR1121, LR2021, LR2022, SX1261, SX1262, or SX1268 connected via SPI/GPIO.

Linux (Raspberry Pi)

ITEM	REQUIREMENT
Host platform	Raspberry Pi with SPI enabled
SPI interface	`/dev/spidev0.0`
GPIO interface	`/dev/gpiochip0`
User groups	User must be in the `spi` and `gpio` groups
Build tools	GCC 12+, CMake 3.25+, Ninja

Baremetal (STM32 / RA0E2)

ITEM	REQUIREMENT
Evaluation board	NUCLEO-L476RG, NUCLEO-L073RZ, or FPB-RA0E2
Build tools	ARM GCC toolchain (`arm-none-eabi-gcc` 12+), CMake 3.25+, Ninja
Serial terminal	ANSI-capable emulator (e.g. Tera Term, PuTTY, minicom) at 115200 baud

1.2 First Test: CW in Three Commands

```
region us
power 22
start cw --freq 915.0
```

The radio is now transmitting an unmodulated carrier at 915 MHz, +22 dBm. Type `stop` or press **Ctrl+C** to halt.

What just happened: `region us` loaded US915 defaults. `power 22` set the PA. `start cw` started transmission with a one-shot frequency override. No modulation setup is needed for CW.

> The CLI saves history to `.radio_test_history`. Use **Up/Down arrows** to recall previous commands. Press **Tab** to complete commands and parameter values.

PART 2

Test Procedures

Step-by-step guides for each test scenario — from CW power measurements to FCC pre-compliance and PER link tests. Each section explains purpose, parameter choices, and what to expect. For complete parameter syntax, see Part 3.

2.1 CW / Conducted Power Measurement

Purpose: Measure conducted output power and verify frequency accuracy. Also used to confirm PA configuration before modulated testing.

Minimum setup

```
region us
power 22
start cw --freq 915.0
```

No modulation is required for CW. The radio transmits a pure carrier at the specified frequency and power.

What to adjust

PARAMETER	HOW TO SET IT	WHY
`freq`	Band edges and center (e.g. 902.3, 915.0, 927.5 MHz)	Verify power flatness across the band
`power`	Step through the power range	Confirm power accuracy at each level
`pa` `pa_ramp_us`	`pa pa_ramp_us 16`	Faster ramp; observe switching transients on the analyzer

Inline power sweep (scripted)

```
region us
start cw --freq 915.0 --power 22
delay 10000
stop
start cw --freq 915.0 --power 14
delay 10000
stop
start cw --freq 915.0 --power 0
delay 10000
stop
exit
```

> `--freq` and `--power` on `start` are one-shot overrides. They do not change the stored configuration, so you can sweep without reconfiguring between runs.

2.2 FCC FHSS Test — 64-Channel Hopping

Purpose: Demonstrate compliance with FCC §15.247(a)(1) — frequency hopping spread spectrum. The radio hops across all 64 US915 uplink channels at the configured data rate, proving the system uses at least 50 hopping frequencies with a channel occupancy time below 400 ms.

Minimum setup

```
region us
modulation lora
start fhss --dr 0 --count 0
```

This is a complete, valid FHSS test. The CLI loads LoRa defaults for DR0 (SF10/125 kHz), computes the maximum compliant payload, and hops indefinitely through all 64 channels.

Which DR to use

DR	SF	BW	NOTES
DR0	SF10	125 kHz	Most commonly requested by test labs; longest airtime per hop
DR3	SF7	125 kHz	Shortest airtime at 125 kHz; useful for rapid channel cycling
DR4	SF8	500 kHz	500 kHz bandwidth; different PA path on some chips

For FCC pre-compliance the test lab will specify which DR to use. DR0 is most commonly requested.

Dwell time

At DR0 (SF10/125 kHz) with the maximum auto-computed payload, each hop occupies approximately 370 ms — safely under the FCC 400 ms limit. The CLI warns if a manual payload size would exceed this limit.

Adding inter-hop delay

```
start fhss --dr 0 --count 64 --delay 100
```

The `--delay` option adds a pause (in milliseconds) between hops. This gives the spectrum analyzer time to settle and capture each channel cleanly. A delay of 50–100 ms is typical for swept-tuned analyzers.

Verifying the channel plan

```
show channels
```

Lists all 64 US915 uplink channels with their index and frequency. Use this to verify the channel plan matches your test plan before starting.

2.3 FCC DTS Test — 500 kHz Digital Modulation

Purpose: Demonstrate compliance with FCC §15.247(a)(3) — digital transmission system (DTS). The radio transmits continuously on a single frequency using 500 kHz LoRa bandwidth, proving the modulated signal meets spectral mask and power requirements without frequency hopping.

Setup

```
region us
modulation lora
bw 500
sf 7
cr 4/5
pld auto
start dts --freq 915.0
```

Why these parameters:

- `bw 500` — 500 kHz is required for DTS qualification under §15.247(a)(3)
- `sf 7` — lowest spreading factor gives the widest occupied bandwidth, stressing the spectral mask
- `cr 4/5` — minimum coding rate (LoRaWAN standard)
- `pld auto` — maximum payload; no dwell-time limit applies at 500 kHz

> No dwell-time limit applies to 500 kHz DTS operation.

Varying the test frequency

```
start dts --freq 903.0
start dts --freq 915.0
start dts --freq 927.0
```

2.4 FCC Hybrid Test — 8-Channel Sub-Band

Purpose: Demonstrate compliance for systems that hop across fewer than 50 channels within an 8-channel sub-band of the US915 plan. The radio hops through 8 consecutive channels using 500 kHz LoRa bandwidth.

Setup

```
region us
modulation lora
bw 500
sf 7
start hybrid --mask 0 --dr 4 --count 0
```

Sub-band mask reference

MASK	CHANNELS	FREQUENCY RANGE
0	ch 0–7	902.3–903.7 MHz
1	ch 8–15	903.9–905.3 MHz
2	ch 16–23	905.5–906.9 MHz
3	ch 24–31	907.1–908.5 MHz
4	ch 32–39	908.7–910.1 MHz
5	ch 40–47	910.3–911.7 MHz
6	ch 48–55	911.9–913.3 MHz
7	ch 56–63	913.5–914.9 MHz

DR4 (SF8/500 kHz) is the standard LoRaWAN data rate for 500 kHz channels and is the natural choice for hybrid testing.

Sweeping all sub-bands (script)

```
start hybrid --mask 0 --dr 4 --count 8 --delay 100
start hybrid --mask 1 --dr 4 --count 8 --delay 100
start hybrid --mask 2 --dr 4 --count 8 --delay 100
start hybrid --mask 3 --dr 4 --count 8 --delay 100
start hybrid --mask 4 --dr 4 --count 8 --delay 100
start hybrid --mask 5 --dr 4 --count 8 --delay 100
start hybrid --mask 6 --dr 4 --count 8 --delay 100
start hybrid --mask 7 --dr 4 --count 8 --delay 100
```

2.5 PER Link Test

Purpose: Measure packet error rate between two radio units to verify link budget, sensitivity, and antenna performance. One unit transmits a known number of packets; the other counts received packets and reports RSSI, SNR, and PER.

Setup — Unit A (transmitter)

```
region us
modulation lora
freq 915.0
power 14
bw 125
sf 10
cr 4/5
per count 1000
per interval 200
per payload 16
start per-tx
```

Setup — Unit B (receiver)

Configure identically (region, modulation, freq, bw, sf, cr, syncword, header, crc), then:

```
region us
modulation lora
freq 915.0
bw 125
sf 10
cr 4/5
start per-rx
```

> Start the receiver before the transmitter. The receiver must be listening before the first packet is sent, or the initial packets will be lost and inflate the PER result.

Reading results

```
per stats
```

```
Last PER test result (rx):
  Received:   997
  CRC errors:   0
  PER:        0.30%
  RSSI avg:  -78 dBm
  SNR avg:   +9.2 dB
```

Sensitivity sweep

Reduce TX power in steps to find the sensitivity threshold where PER degrades:

```
start per-tx --power 14
start per-tx --power 10
start per-tx --power 5
start per-tx --power 0
start per-tx --power -5
```

2.6 FLRC Test (LR2021)

Purpose: Test the FLRC (Fast Long Range Communication) modulation mode available on the LR2021 radio. FLRC provides higher data rates (up to 2600 kbps). Available on LR2021 only, in `us` and `2g4` regions.

Conducted power / spectral mask

```
region 2g4
modulation flrc
br 1300
cr none
bt bt0.5
preamble 32
sw_len 4
header variable
crc 2
pld 127
start modulated --freq 2440.0 --power 10
```

Parameter guidance:

- `bt bt0.5` — Gaussian BT product of 0.5 gives a good balance of spectral efficiency and ISI; tighter spectral mask
- `bt off` — disables Gaussian filtering for maximum data throughput at the cost of wider spectral occupancy
- `cr none` — no coding rate overhead; use `1/2` or `3/4` for forward error correction in noisy environments

FLRC PER link test

Unit A (TX):

```
region 2g4
modulation flrc
br 1300
cr none
bt bt0.5
preamble 32
sw_len 4
syncword ED592398
header variable
crc 2
per count 1000
per interval 50
per payload 64
start per-tx --freq 2440.0 --power 10
```

Unit B (RX):

```
region 2g4
modulation flrc
br 1300
cr none
bt bt0.5
preamble 32
sw_len 4
syncword ED592398
header variable
crc 2
start per-rx --freq 2440.0
```

> Both units must use identical `br`, `cr`, `bt`, `preamble`, `sw_len`, `syncword`, `header`, and `crc` settings. The `syncword` value must match between transmitter and receiver.

PART 3

Command Reference

Complete syntax and valid values for every command and parameter. Use this when you need exact details — Part 2 covers when and why to use them.

3.1 Region and Modulation

`region` — Set Regulatory Region

```
region <us|2g4>
```

Sets the regulatory region and loads default frequency and power values. Must be the first command issued after startup.

REGION ID	BAND	FREQUENCY RANGE	MAX POWER	REGULATORY MODES
`us`	US915	902.0–928.0 MHz	+22 dBm	FHSS, DTS, Hybrid
`2g4`	WW-2G4	2400.0–2480.0 MHz	+12 dBm	—

`modulation` — Set Modulation Type

```
modulation <lora|flrc>
modulation help
```

Sets the modulation type and loads region-appropriate parameter defaults. `modulation help` lists all modulations supported by the connected chip.

LoRa is available on all chips. FLRC is available on LR2021 only, in `us` and `2g4` regions.

`freq` — Set Center Frequency

```
freq <MHz>
```

Sets the transmit/receive center frequency in MHz. The value must be within the current region's allowed range.

`power` — Set TX Power

```
power <dBm>
```

Sets the transmit output power in dBm. The value is validated against the current region's minimum and maximum limits.

`agc` — Receiver Gain Control

```
agc <auto|g1|g2|...|g13>
```

Sets the receiver gain mode. `auto` (default) enables automatic gain control. `g1` through `g13` select a fixed gain step (g1 = maximum gain, g13 = minimum gain). Available on LR20xx chips only.

VALUE	DESCRIPTION
`auto`	Automatic gain control (default)
`g1` – `g13`	Fixed gain step (g1 = max gain, g13 = min gain)

`boost-lf` / `boost-hf` — RX Boost Level

```
boost-lf <auto|0-7>
boost-hf <auto|0-7>
```

Sets the RX boost level for the low-frequency (`boost-lf`) or high-frequency (`boost-hf`) receive path. `auto` uses the default value (0 for LF, 4 for HF). Available on LR20xx chips only.

`xosc` — Crystal Oscillator Trimming

```
xosc <show|default|xta|xtb|wait> [value]
```

Manages crystal oscillator trimming capacitor values. Available on LR20xx chips only.

SUBCOMMAND	DESCRIPTION
`xosc show`	Display current XTA, XTB, and wait values (with pF conversion)
`xosc default`	Restore BSP default trim values
`xosc xta <0-47>`	Set XTA capacitor trim (11.3 + N * 0.47 pF)
`xosc xtb <0-47>`	Set XTB capacitor trim (11.1 + N * 0.47 pF)
`xosc wait <0-255>`	Set stabilization delay in microseconds

`status` — Show Current Configuration

```
status
```

Displays all current configuration parameters in a single view, including region, modulation, frequency, power, modulation-specific parameters, PA overrides, and test mode state.

3.2 LoRa Parameters

LoRa parameters are set by typing the parameter name directly at the prompt after `modulation lora` is active. All parameters are validated against regional constraints before being applied.

`bw` — Bandwidth

```
bw <kHz>
```

REGION	VALID VALUES (KHZ)
US915	125, 250, 500
WW-2G4	812

> Changing bandwidth may invalidate the current spreading factor. The CLI warns when this occurs; use `sf` to adjust before starting a test.

`sf` — Spreading Factor

```
sf <5-12>
```

BW (KHZ)	US915 SF RANGE	LORAWAN DR EQUIVALENT
125	SF5–SF10	DR0 (SF10) – DR3 (SF7)
250	SF5–SF12	—
500	SF5–SF12	DR4 (SF8/500k)

`cr` — Coding Rate

```
cr <4/5|4/6|4/7|4/8> — all chips
cr <li4/5|li4/6|li4/8> — LR11xx, LR20xx only
cr <lic4/6|lic4/8> — LR20xx only
```

Sets the LoRa coding rate. Standard rates `4/5` through `4/8` are available on all chips. Long interleaved (LI) rates are available on LR11xx and LR20xx. Long interleaved convolutional (LI-Conv) rates are LR20xx only.

VALUE	TYPE	CHIPS
`4/5`, `4/6`, `4/7`, `4/8`	Standard	All
`li4/5`, `li4/6`, `li4/8`	Long interleaved	LR11xx, LR20xx
`lic4/6`, `lic4/8`	Long interleaved convolutional	LR20xx only

`preamble` — Preamble Length

```
preamble <4–65535>
```

Sets the LoRa preamble length in symbols. Default is 8 symbols (LoRaWAN standard). Longer preambles improve reception reliability at the cost of airtime.

`syncword` — Sync Word

```
syncword <hex>
```

Sets the LoRa sync word as a one-byte hexadecimal value.

VALUE	MEANING
`0x34`	LoRaWAN public network (default for US)
`0x12`	Private network
`0x21`	WW-2G4 public (default for 2g4)

`header` — Header Mode

```
header <explicit|implicit>
```

In **explicit** mode (default), the packet header carries the payload length, coding rate, and CRC presence flag. In **implicit** mode, no header is transmitted — both TX and RX must agree on the payload length and CRC setting in advance.

`crc` — CRC Enable

```
crc <on|off>
```

Enables or disables the LoRa payload CRC. Default is `on`.

`iq` — I/Q Polarity

```
iq <standard|inverted>
```

Sets the I/Q polarity. `standard` (default) uses normal I/Q mapping. `inverted` swaps I and Q, which is used by some LoRaWAN downlink configurations. Available on all chips.

`pld` — Payload Size

```
pld <1–255|auto>
```

Sets the payload size in bytes for continuous modulated transmission (`start modulated`). Setting `auto` computes the maximum payload that fits within the FCC 400 ms dwell-time limit for the current LoRa configuration.

> **FCC dwell time:** At 125 kHz and 250 kHz bandwidths, the FCC limits each transmission to 400 ms per channel before hopping is required. The CLI calculates time-on-air and warns if the configured payload exceeds this limit. 500 kHz (DTS) is exempt from the dwell-time rule.

`ldro` — Low Data Rate Optimization

```
ldro <on|off|auto>
```

Controls the low data rate optimization flag. When set to `auto` (default), the CLI enables LDRO automatically when the symbol duration exceeds 16 ms (SF11/125 kHz and SF12/125 kHz). Force `on` or `off` to override. Available on SX126x and LR11xx only (not available on LR20xx).

3.3 FLRC Parameters

FLRC parameters are available after `modulation flrc` is active. FLRC is supported on LR2021 only, in `us` and `2g4` regions.

`br` — Bit Rate

```
br <260|325|520|650|1040|1300|2080|2600>
```

Sets the FLRC bit rate in kbps. Higher bit rates reduce airtime but require better link budget.

VALUE (KBPS)	BANDWIDTH
260	312 kHz
325	312 kHz
520	624 kHz
650	624 kHz
1040	1248 kHz
1300	1248 kHz
2080	2496 kHz
2600	2496 kHz

`cr` — Coding Rate

```
cr <1/2|2/3|3/4|none>
```

Sets the FLRC forward error correction coding rate. `none` disables FEC for maximum throughput. Default is `3/4`.

`bt` — Gaussian BT Filter

```
bt <off|bt0.5|bt1>
```

Controls the Gaussian pulse-shaping filter. `bt0.5` gives tighter spectral mask (recommended). `bt1` provides moderate filtering. `off` disables filtering for maximum throughput. Default is `bt0.5`.

`preamble` — Preamble Length

```
preamble <4|8|12|16|20|24|28|32>
```

Sets the FLRC preamble length in bits. Must be a multiple of 4 from 4 to 32.

`sw_len` — Sync Word Length

```
sw_len <off|2|4>
```

Sets the sync word length in bytes. `off` disables the sync word. Default is `4`.

`tx_sw` — TX Sync Word Index

```
tx_sw <off|1|2|3>
```

Selects which syncword register to transmit with. `off` disables syncword transmission. Default is `1`.

`rx_sw` — RX Sync Word Match

```
rx_sw <off|1-7>
```

Selects which syncword register(s) to match on receive, as a bitmask. Default is `1` (syncword 1 only).

VALUE	MATCHES
`off` / `0`	None (syncword matching disabled)
`1`	sw1
`2`	sw2
`3`	sw1 | sw2
`4`	sw3
`5`	sw1 | sw3
`6`	sw2 | sw3
`7`	sw1 | sw2 | sw3

`header` — Header Mode

```
header <variable|fixed>
```

`variable` (default) includes the payload length in the packet header. `fixed` omits the length header — both TX and RX must agree on the payload length.

`crc` — CRC Length

```
crc <off|2|3|4>
```

Sets the CRC length in bytes. `off` disables CRC. Default is `2`.

`syncword` — Sync Word Value

```
syncword <hex>
```

Sets the 4-byte syncword value for register 1. Enter 1 to 8 hex digits; values shorter than 8 digits are zero-padded from the left. Default is `0xED592398`.

> **FLRC syncword architecture:** `syncword` sets the byte values in the syncword register. `tx_sw` selects which register to transmit with. `rx_sw` controls which registers are matched on receive. For basic operation, set `syncword` to the desired value and leave `tx_sw` and `rx_sw` at their defaults (both `1`).

3.4 PA Configuration

The power amplifier is automatically configured when the `power` command is used. For advanced measurements requiring manual PA control, use the `pa` command.

```
pa show
pa reset
pa <param> <value>
```

`pa show` displays the current PA register values and valid ranges. `pa reset` clears all manual overrides and restores the automatic PA configuration for the current power level. `pa <param> <value>` sets an individual PA register.

Override semantics:

- Manual PA overrides take effect immediately and persist until `pa reset` is called or the `power` command is used
- The `power` command recalculates all PA registers from the target power level, clearing any manual overrides
- Overrides are not saved between sessions

LR20xx (LR2021, LR2022)

PARAMETER	RANGE	DESCRIPTION
`half_power`	−39 to 44	TX power in 0.5 dB steps
`pa_sel`	0–1 (read-only)	PA path: 0 = LF, 1 = HF
`pa_lf_duty_cycle`	0–31	Low-freq PA duty cycle
`pa_lf_slices`	0–7	Low-freq PA number of slices
`pa_hf_duty_cycle`	16–31	HF duty cycle (16 = max, 31 = lowest)
`pa_ramp_us`	2–304	PA ramp time in microseconds

> `pa_sel` is read-only and set automatically based on frequency: LF PA for sub-GHz, HF PA for 2.4 GHz. `pa_ramp_us` accepts discrete values only: 2, 4, 8, 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 240, 272, 304.

SX1262 / SX1268

PARAMETER	RANGE	DESCRIPTION
`power`	−9 to 22	TX power register value (dBm)
`pa_duty_cycle`	0–4	PA duty cycle (max 0x04, see datasheet)
`hp_max`	0–7	HP PA limit (7 = max, enables +22 dBm)
`pa_ramp_us`	10–3400	PA ramp time in microseconds

SX1261

PARAMETER	RANGE	DESCRIPTION
`power`	−17 to 14	TX power register value (dBm)
`pa_duty_cycle`	0–7	PA duty cycle (see datasheet for max)
`pa_ramp_us`	10–3400	PA ramp time in microseconds

LR11xx (LR1110, LR1120, LR1121)

PARAMETER	RANGE	DESCRIPTION
`power`	−17 to 22	TX power in dBm
`pa_sel`	0–2 (read-only)	PA path: 0 = LP, 1 = HP, 2 = HF
`pa_reg_supply`	0–1 (read-only)	PA supply: 0 = VREG, 1 = VBAT
`pa_duty_cycle`	0–7	PA duty cycle
`pa_hp_sel`	0–7	HP PA slices
`pa_ramp_us`	16–304	PA ramp time in microseconds

> `pa_sel` and `pa_reg_supply` are read-only — the BSP selects LP/HP/HF based on frequency and power level. `pa_ramp_us` accepts discrete values only: 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 240, 272, 304.

3.5 Test Modes and Inline Overrides

Test modes are started with `start <mode>` and stopped with `stop` or **Ctrl+C**. Only one mode may be active at a time. Configuration commands are blocked while any mode is running.

> On baremetal MCU targets, the CLI runs in a polled main loop. `stop` is the primary way to halt a test. **Ctrl+C** is handled on Linux only.

`start cw` — Unmodulated Carrier

Transmits a continuous single-frequency unmodulated carrier. No modulation needs to be configured — only frequency and power are required.

```
start cw [--freq <MHz>] [--power <dBm>]
```

`start modulated` — Continuous Modulated Burst

Transmits packets back-to-back using the current modulation configuration. Used for occupied bandwidth measurements and spectral mask tests. Payload size is set with the `pld` command.

```
start modulated [--freq <MHz>] [--power <dBm>]
```

`start rx` — Continuous Receive

Places the radio in continuous receive mode. Each received packet is printed with RSSI and SNR. CRC errors are reported separately.

```
start rx [--freq <MHz>]
```

`start fhss` — Frequency Hopping (US only)

Transmits using the US915 64-channel FHSS plan. FCC §15.247(a)(1) pre-compliance.

```
start fhss [--dr <0-6>] [--count <N>] [--delay <ms>] [--power <dBm>]
```

`start dts` — Digital Transmission System (US only)

Transmits continuously on a fixed frequency at 500 kHz bandwidth. FCC §15.247(a)(3) pre-compliance.

```
start dts [--freq <MHz>] [--power <dBm>] [--count <N>] [--delay <ms>]
```

`start hybrid` — 8-Channel Sub-Band (US only)

Hops through 8 consecutive channels within a selectable sub-band. The `--mask` argument is required.

```
start hybrid --mask <0-7> [--dr <0-6>] [--count <N>] [--delay <ms>] [--power <dBm>]
```

`start per-tx` / `start per-rx` — Packet Error Rate

Runs a PER test between two radio units. Configure PER parameters with the `per` command before starting.

```
start per-tx [--freq <MHz>] [--power <dBm>]
start per-rx [--freq <MHz>]
```

`stop` — Stop Active Test

Stops the currently running test mode and returns to idle.

```
stop
```

Inline overrides

The `start` command accepts inline overrides that modify parameters for a single test run without changing the stored configuration.

OVERRIDE	APPLICABLE MODES	DESCRIPTION
`--freq <MHz>`	All	One-shot frequency override
`--power <dBm>`	All TX modes	One-shot power override
`--dr <0-6>`	fhss, hybrid	LoRaWAN data rate for this run
`--count <N>`	fhss, dts, hybrid	Packet/hop count (0 = infinite)
`--delay <ms>`	fhss, dts, hybrid	Inter-packet/hop delay
`--mask <0-7>`	hybrid (required)	8-channel sub-band selection

3.6 PER Configuration

COMMAND	DESCRIPTION
`per count <N>`	Packets TX will send; 0 = run indefinitely (default: 100)
`per interval <ms>`	Inter-packet gap on TX side; must be > 0 (default: 500 ms)
`per payload <6-255|auto>`	Payload size per packet; `auto` = max for FCC dwell time (default: auto)
`per stats`	Display last test result (available any time, even during a test)
`per reset`	Clear last saved result

> `per stats` and `per reset` are always available. All other `per` subcommands require the radio to be idle — use `stop` first.

3.7 Utility Commands and Scripting

`show channels`

Displays the complete LoRaWAN channel plan for the current region, listing each channel index and its center frequency.

```
show channels
```

`delay`

Pauses execution for the specified duration in milliseconds. If a test mode is active, it continues to run during the delay — this is intended for use in scripts to set observation windows.

```
delay <ms>
```

`term` — Terminal Input Mode

```
term <ansi|plain>
```

Switches the terminal input mode between `ansi` (arrow keys, history, tab completion) and `plain` (basic line input). MCU only — not available on Linux. With no arguments, shows the current mode.

`version`

Displays the CLI version string and build information.

```
version
```

`help`

Shows the command reference. Provide a command name for context-sensitive detail.

```
help [command]
```

`exit` / `quit`

Stops any active test, closes the radio, and exits the CLI.

```
exit
```

Scripting / Pipe Mode

When stdin is not a terminal (i.e. piped or redirected input), the CLI runs in non-interactive script mode: commands are read sequentially from stdin and the CLI exits when EOF is reached.

```
./build/radio_test_cli <<'EOF'
region us
modulation lora
freq 915.0
power 22
start cw
delay 30000
stop
exit
EOF
```

> Commands that return an error do not stop script execution. Output is written to stdout in the same format as interactive mode, making it suitable for capture with `tee` or redirection.

PART 4

Reference Tables

Region parameters, chip feature matrix, and the complete command quick-reference.

4.1 Region Reference

US915

PARAMETER	VALUE
Frequency range	902.0–928.0 MHz
Default frequency	902.3 MHz (LoRaWAN ch0)
TX power range	−10 to +22 dBm (LR20xx); −9 to +22 dBm (SX1262/SX1268); −17 to +22 dBm (SX1261, LR11xx)
Default TX power	+22 dBm
LoRa BW options	125, 250, 500 kHz
LoRa SF (125 kHz)	SF5–SF10
LoRa SF (250/500 kHz)	SF5–SF12
Default SF/BW	SF10 / 125 kHz (DR0)
Default sync word	0x34 (LoRaWAN public)
Regulatory modes	FHSS (64-ch), DTS (500 kHz), Hybrid (8-ch sub-band)

US915 Data Rates

DR	MODULATION	SF	BW	BIT RATE (APPROX)
DR0	LoRa	SF10	125 kHz	980 bps
DR1	LoRa	SF9	125 kHz	1760 bps
DR2	LoRa	SF8	125 kHz	3125 bps
DR3	LoRa	SF7	125 kHz	5470 bps
DR4	LoRa	SF8	500 kHz	12500 bps

WW-2G4 (LR2021, LR2022, LR1120, LR1121)

PARAMETER	VALUE
Frequency range	2400.0–2480.0 MHz
Default frequency	2423.0 MHz

PARAMETER	VALUE
TX power range	−17 to +12 dBm (LR20xx); −18 to +12 dBm (LR11xx)
Default TX power	+10 dBm
LoRa BW	812 kHz
LoRa SF range	SF5–SF12
Default sync word	0x21 (WW-2G4 public)
FLRC	Available (LR2021 only)
Regulatory modes	None (handled by host protocol stack)

4.2 Chip Compatibility

FEATURE	LR2021	LR2022	LR1110	LR1120	LR1121	SX1261	SX1262	SX1268
LoRa (Sub-GHz)	✓	✓	✓	✓	✓	✓	✓	✓
LoRa (2.4 GHz)	✓	✓	—	✓	✓	—	—	—
FLRC	✓	—	—	—	—	—	—	—
Region: us	✓	✓	✓	✓	✓	✓	✓	✓
Region: 2g4	✓	✓	—	✓	✓	—	—	—
FHSS / DTS / Hybrid	✓	✓	✓	✓	✓	✓	✓	✓
PER TX/RX	✓	✓	✓	✓	✓	✓	✓	✓
AGC control	✓	✓	—	—	—	—	—	—
RX boost (LF/HF)	✓	✓	—	—	—	—	—	—
XOSC trimming	✓	✓	—	—	—	—	—	—
LED indicators	—	—	✓	✓	✓	—	—	—
I/Q polarity	✓	✓	✓	✓	✓	✓	✓	✓
LDRO control	—	—	✓	✓	✓	✓	✓	✓
CR LI/LI-Conv	✓	✓	✓	✓	✓	—	—	—

4.3 Quick-Reference Command Table

COMMAND	SYNTAX	DESCRIPTION
`region`	`region <us|2g4>`	Set region; load defaults
`modulation`	`modulation <lora|flrc>`	Set modulation; load defaults
`freq`	`freq <MHz>`	Set center frequency
`power`	`power <dBm>`	Set TX power
`agc`	`agc <auto|g1-g13>`	Receiver gain control (LR20xx only)
`boost-lf`	`boost-lf <auto|0-7>`	RX boost level, LF path (LR20xx only)
`boost-hf`	`boost-hf <auto|0-7>`	RX boost level, HF path (LR20xx only)
`xosc`	`xosc <show|default|xta|xtb|wait>`	Crystal oscillator trimming (LR20xx only)
`status`	`status`	Show complete current configuration
`bw`	`bw <kHz>`	LoRa bandwidth (125/250/500/812)
`sf`	`sf <5-12>`	LoRa spreading factor
`cr`	`cr <4/5|4/6|4/7|4/8|li4/5|li4/6|li4/8|lic4/6|lic4/8>`	LoRa coding rate
`preamble`	`preamble <symbols>`	Preamble length
`syncword`	`syncword <hex>`	LoRa sync word (e.g. 0x34)
`header`	`header <explicit|implicit>`	LoRa header mode
`crc`	`crc <on|off>`	LoRa CRC enable
`iq`	`iq <standard|inverted>`	LoRa I/Q polarity
`pld`	`pld <1-255|auto>`	Modulated TX payload size
`ldro`	`ldro <on|off|auto>`	Low data rate optimization (SX126x, LR11xx only)
`br`	`br <260|325|520|650|1040|1300|2080|2600>`	FLRC bit rate (kbps)
`cr` (FLRC)	`cr <1/2|2/3|3/4|none>`	FLRC coding rate
`bt`	`bt <off|bt0.5|bt1>`	FLRC Gaussian BT filter

<table>
<tr><th>COMMAND</th><th>SYNTAX</th><th>DESCRIPTION</th></tr>
<tr><td><code>preamble</code> (FLRC)</td><td><code>preamble <4|8|12|16|20|24|28|32></code></td><td>FLRC preamble length (bits)</td></tr>
<tr><td><code>sw_len</code></td><td><code>sw_len <off|2|4></code></td><td>FLRC sync word length (bytes)</td></tr>
<tr><td><code>tx_sw</code></td><td><code>tx_sw <off|1|2|3></code></td><td>FLRC TX sync word index</td></tr>
<tr><td><code>rx_sw</code></td><td><code>rx_sw <off|1-7></code></td><td>FLRC RX sync word match (bitmask)</td></tr>
<tr><td><code>header</code> (FLRC)</td><td><code>header <variable|fixed></code></td><td>FLRC header type</td></tr>
<tr><td><code>crc</code> (FLRC)</td><td><code>crc <off|2|3|4></code></td><td>FLRC CRC length (bytes)</td></tr>
<tr><td><code>syncword</code> (FLRC)</td><td><code>syncword <hex></code></td><td>FLRC sync word value (e.g. ED592398)</td></tr>
<tr><td><code>pa show</code></td><td><code>pa show</code></td><td>Display PA parameters and ranges</td></tr>
<tr><td><code>pa reset</code></td><td><code>pa reset</code></td><td>Clear PA overrides</td></tr>
<tr><td><code>pa <p> <v></code></td><td><code>pa <param> <value></code></td><td>Set PA parameter manually</td></tr>
<tr><td><code>start cw</code></td><td><code>start cw [overrides]</code></td><td>Unmodulated carrier TX</td></tr>
<tr><td><code>start modulated</code></td><td><code>start modulated [overrides]</code></td><td>Continuous modulated TX</td></tr>
<tr><td><code>start rx</code></td><td><code>start rx [overrides]</code></td><td>Continuous receive mode</td></tr>
<tr><td><code>start fhss</code></td><td><code>start fhss [overrides]</code></td><td>64-channel FHSS TX (US only)</td></tr>
<tr><td><code>start dts</code></td><td><code>start dts [overrides]</code></td><td>DTS 500 kHz TX (US only)</td></tr>
<tr><td><code>start hybrid</code></td><td><code>start hybrid --mask N [overrides]</code></td><td>Hybrid 8-ch TX (US only)</td></tr>
<tr><td><code>start per-tx</code></td><td><code>start per-tx [overrides]</code></td><td>PER test transmitter</td></tr>
<tr><td><code>start per-rx</code></td><td><code>start per-rx [overrides]</code></td><td>PER test receiver</td></tr>
<tr><td><code>stop</code></td><td><code>stop</code></td><td>Stop active test mode</td></tr>
<tr><td><code>per count</code></td><td><code>per count <N></code></td><td>PER packet count (0 = infinite)</td></tr>
<tr><td><code>per interval</code></td><td><code>per interval <ms></code></td><td>PER inter-packet gap</td></tr>
<tr><td><code>per payload</code></td><td><code>per payload <6-255|auto></code></td><td>PER payload size</td></tr>
<tr><td><code>per stats</code></td><td><code>per stats</code></td><td>Show last PER test result</td></tr>
<tr><td><code>per reset</code></td><td><code>per reset</code></td><td>Clear last PER result</td></tr>
<tr><td><code>show channels</code></td><td><code>show channels</code></td><td>Display region channel plan</td></tr>
</table>

COMMAND	SYNTAX	DESCRIPTION
`delay`	`delay <ms>`	Pause execution (scripting)
`term`	`term <ansi|plain>`	Terminal input mode (MCU only)
`version`	`version`	Show CLI version
`help`	`help [command]`	Show help
`exit` / `quit`	`exit`	Exit CLI

PART 5

Troubleshooting

Common errors and how to resolve them.

Troubleshooting

SPI device not found

Symptom: `Failed to open /dev/spidev0.0` on startup.

Fix: Enable SPI on the Raspberry Pi. Run `sudo raspi-config`, navigate to Interface Options > SPI, and enable it. Reboot. Verify with `ls /dev/spidev*`.

Permission denied on SPI or GPIO

Symptom: `Permission denied` when accessing `/dev/spidev0.0` or `/dev/gpiochip0`.

Fix: Add your user to the required groups:

```
sudo usermod -aG spi,gpio $USER
```

Log out and back in for the group changes to take effect.

Build fails on GCC 14

Symptom: Compilation error about incompatible pointer types in vendor code.

Fix: Add the warning suppression flag:

```
env CFLAGS="-Wno-incompatible-pointer-types" \
  cmake -UCMAKE_C_FLAGS -S examples -B build -DBOARD=LINUX -DRAC_RADIO=lr2021 -G Ninja
```

Radio not responding / no RX packets

Symptom: Commands succeed but no signal is observed on the spectrum analyzer, or the receiver sees no packets.

Fix: Check the following:

- Verify the SPI connection and that the correct radio shield is seated properly
- Confirm the `-DRAC_RADIO=` build flag matches the actual radio chip connected
- For PER tests, ensure both units have identical modulation parameters (bw, sf, cr, syncword, header, crc)
- Check that the frequency is within the selected region's valid range

FLRC: "not supported by this chip"

Symptom: `modulation flrc` returns an error.

Fix: FLRC is available on LR2021 only. Verify the build was compiled with `-DRAC_RADIO=lr2021`. FLRC is available in both `us` and `2g4` regions.

Region 2g4: "not supported by this chip"

Symptom: `region 2g4` returns an error on SX126x or LR11xx.

Fix: The 2.4 GHz region is only available on chips with a 2.4 GHz radio path: LR2021, LR2022, LR1120, and LR1121. SX1261, SX1262, SX1268, and LR1110 are sub-GHz only.

AGC command not recognized

Symptom: `agc auto` returns an error.

Fix: The `agc` command is available on LR20xx chips only (LR2021, LR2022). It is not supported on SX126x or LR11xx hardware.
